Date of issue	2024-09-19
Version	1.0
Total pages	22

Test report of
IES LM-79-08
Approved Method: Electrical and Photometric Measurements of Solid-State Lighting Products

Applicant:

RZ Lighting Co Ltd

Address:

4F, Bldg. B, Zhongjing Sci-Tech Park 606 Fengtang Blvd, Bao'an Shenzhen, Guangdong Sheng 518103 China

For Product:

Outdoor Non-Cutoff and Semi-Cutoff Wall-Mounted Area Luminaires

Model No.:

WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01

Test laboratory: Shenzhen Belling Efficiency Testing Lab Co., Ltd, 1Floor, No.1 Building, Meibaohe Industrial Park, Dalang Street, Longhua District, Shenzhen, Guangdong Prov.518101 China.

Complied by: Sam Chen	**Review by: Jason zhou**
Project Engineer	**Technical Manager**

Note 1: The test data was only valid for the test sample(s). This test report is prepared for the customer shown above and for the device described herein. It may not be duplicated or used in part without prior written consent from Shenzhen Belling Efficiency Testing Lab Co., Ltd. This report must not be used by the customer to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the U.S. Government.
Note 2: This report contains data that are not covered by the NVLAP accreditation. See the following description: TM-15-11 test are not in NVLAP accreditation scope.

1 General

1.1 Product Information

Manufacturer	RZ Lighting Co Ltd
Manufacturer Address	4F, Bldg. B, Zhongjing Sci-Tech Park 606 Fengtang Blvd, Bao'an Shenzhen, Guangdong Sheng 518103 China
Brand Name	RZ
Luminaire Type	Outdoor Non-Cutoff and Semi-Cutoff Wall-Mounted Area Luminaires
Model Number	WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01
Rated Inputs	AC 120-277V, 50/60Hz
Field-Adjustable Product	Yes, Wattage setting: 60W, 90W, 100W, 120W
Color-Tunable Product	Yes, CCT setting: 4000K, 5000K, 5700K
Dimming Capability	Continuous
Integral Control Sensors	Optional
Date of Receipt Samples	2024-09-09
Date of test	2024-09-10 to 2024-09-11
Burning Time Before Test	0hour(For New Products)

1.2 Standards or methods

- ANSI C78.377-2017: Specifications for the Chromaticity of Solid State Lighting Products
- ANSI C82.77-10:2014: Harmonic Emission Limits - Related Power Quality Requirements for Lighting Equipment - Solid State
- CIE Publication No.13.3-1995: Method of Measuring and Specifying Color Rendering of Light Sources
- IESNA LM-79-08 Approved Method: Electric & Photometric Measurement of Solid-state Lighting Products
- IES TM-15-11: Backlight, Uplight, and Glare (BUG) Ratings

1.3 Equipment list

Device	Manufacture	Model No.	Serial No.	Calibration due date
Goniophotometeric System	SENSING	GMS-3000	M101758514120011	2025-04-09
AC Power Source	ALL POWER	ALL POWER	970780	2025-04-17
Total Luminous Flux Standard Lamp	SENSING	110V/100W	S13100188	2025-04-16
Total Luminous Flux Standard Lamp	OSRAM	12V/20W	LSD12201737	2025-04-16
Total Spectral Radiant Flux Standard Lamp	Everfine	D204	M133806CA1411205	2025-04-16
Digital Power Meter	YOKOGAWA	WT310	C2QM02030V	2025-04-17
Thermostatic stabilized photometric sphere	SENSING	SPR-600M	N.A	2025-04-09
Plant spectral photosynthetically radiometer	Everfine	SP-20	P612946CF1411115	2025-04-09
Digital Power Meter	YOKOGAWA	WT210	91L929742	2025-04-17
Spectral radiometer	SENSING	SPR-3000	S1101108	2025-04-09
Environment Measurer	XUYAO	HS-1	N/A	2025-04-11
Environment Measurer	XUYAO	HS-1	N/A	2025-04-11
Stop watch	KISLO	K610	N/A	2025-04-17
Digital Anemometer	TECMAN	TD8901	026141	2024-09-06

Statement of Traceability: Shenzhen Belling Efficiency Testing Lab Co., Ltd attests that all calibration has been performed using suitable standards traceable to national primary standards and International System of Unit (SI).

2 Test conducted and method

2.1 Ambient Condition

The ambient temperature in which measurements are being taken was maintained at 25°C ±1°C, the air flow around the sample(s) being tested did not affect the performance.

2.2 Power Supply Characteristics

The AC power supply had a sinusoidal voltage wave shape at the prescribed frequency (60 Hz) such that the RMS summation of the harmonic components does not exceed 3 percent of the fundamental during operation of the test item.

The voltage of AC power supply (RMS voltage) applied to the device under test was regulated to within±0.2 percent under load.

2.3 Seasoning and Stabilization

No seasoning was performed in accordance with IESNA LM-79-08. And before the measurement, the sample was stabilized until the light output and power variations were less than 0.5% in 30 minutes intervals (3 readings, 15 minutes apart).

2.4 Integrating Sphere System

The system includes AC power source, digital power meter, DC power supply, spectrophotometer, and integrating sphere. The integrating sphere system is calibrated by standard light source before measurement. The system and standard light source has been calibrated regularly and traceable to the National Primary Standards. 4π geometry was used during measurement. The product was operated in its intended orientation in application and was recorded in this report.

Integrating Sphere Uncertainty: The uncertainty of the light output (luminous flux) measurements is U=1.8% (K=2), at the 95% confidence level. The uncertainty of the correlated color temperature measurements is U=20K (K=2), at the 95% confidence level. The uncertainty of the CRI is U=1.8(K=2), at the 95% confidence level. The uncertainty of power meter AC current U=0.18% of rdg, AC Voltage U=0.16% of rdg, Power U=0.20% (K=2), at the 95% confidence level.

2.5 Goniophotometer System

The goniophotometer system is calibrated by standard light source before measurement. The standard light source has been calibrated regularly and traceable to the National Primary Standards.

Type C goniophotometer was used for measuring total luminous flux, luminous intensity distribution, and color spatial uniformity. The product was operated in its intended orientation in application and was recorded in this report. The method according to IESNA LM-79-08 following chapter.

Goniophotometer Uncertainty: The uncertainty of the luminous intensity is U=1.6% (K=2), at the 95% confidence level.

3 Test Result Summary

3.1 Integrating Sphere System (Total operating time for integrating sphere test: 1.0 hour)

3.1.1 Model Number: WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01, 4000K

Electrical data

Input Voltage(V)	Frequency (Hz)	Input Current (A)	Power (W)	Power Factor
120.02	60	0.974	116.72	0.998

Photometric data

Luminous Flux (lm)	Efficacy (lm/W)	CCT (K)
18021.89	154.4	4192

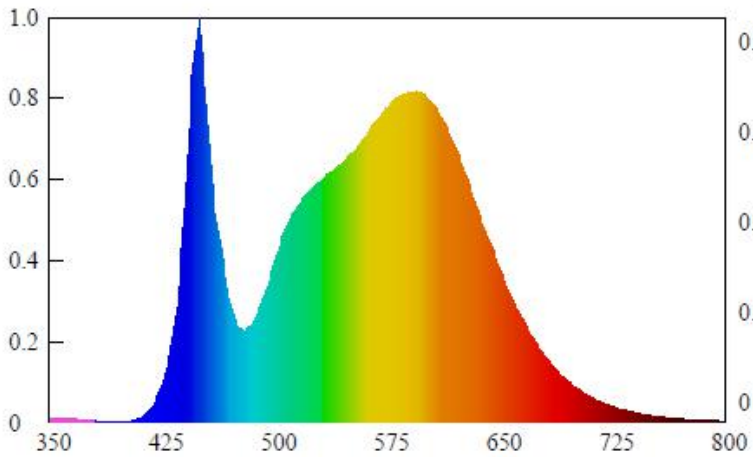
Chromaticity Coordinate

Duv	x	y	u'	v'
-0.0008	0.3719	0.3696	0.2223	0.4971

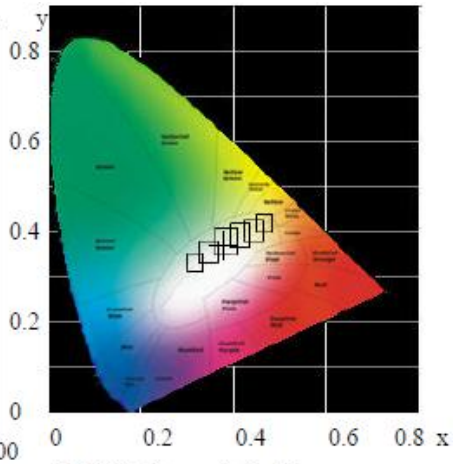
Color Rendering

CRI	R9	Rf	Rg	Rcs,h1(%)
82.2	4	83	96	-13

Spectral Distribution

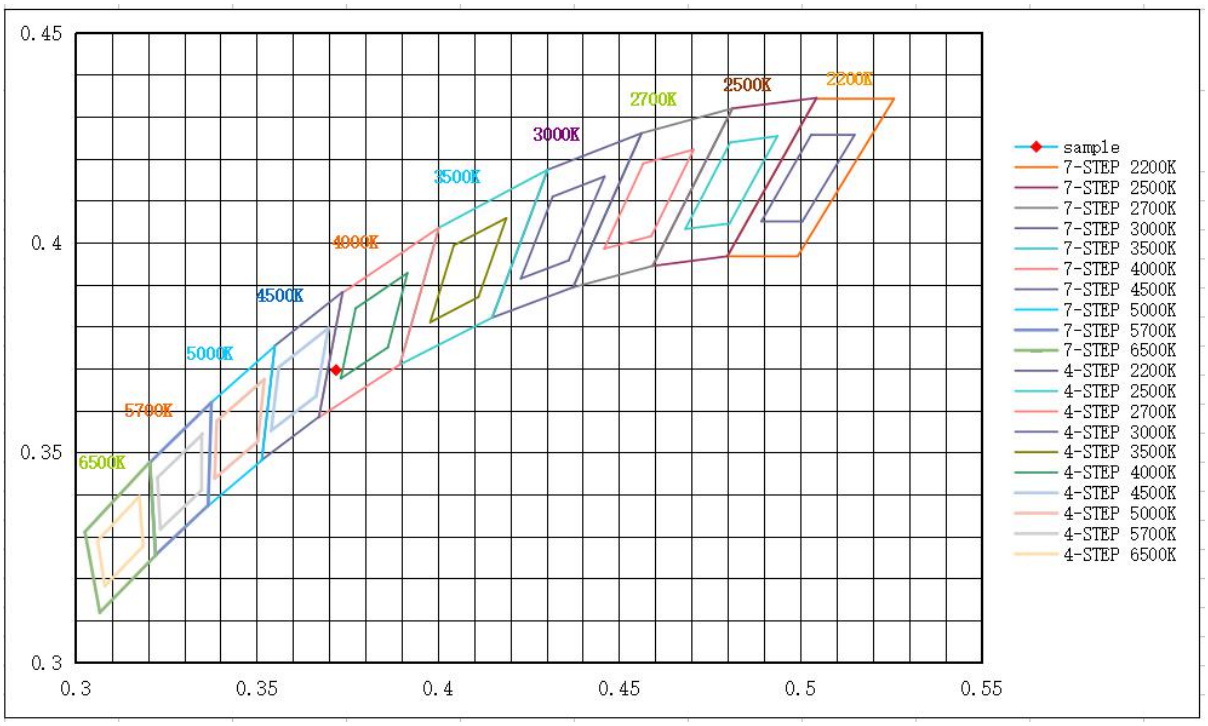


Spectral Distribution



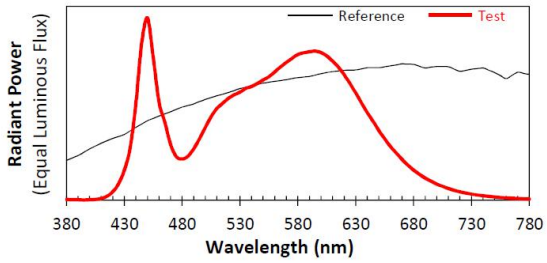
CIE1931 Chromaticity Diagram

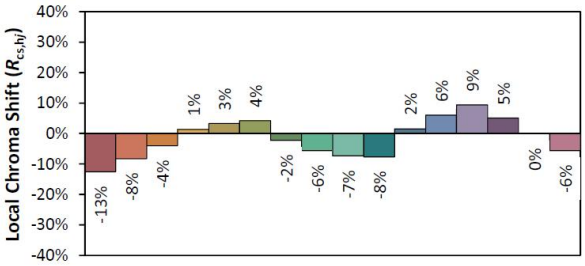
7/4 Step Quadrangle

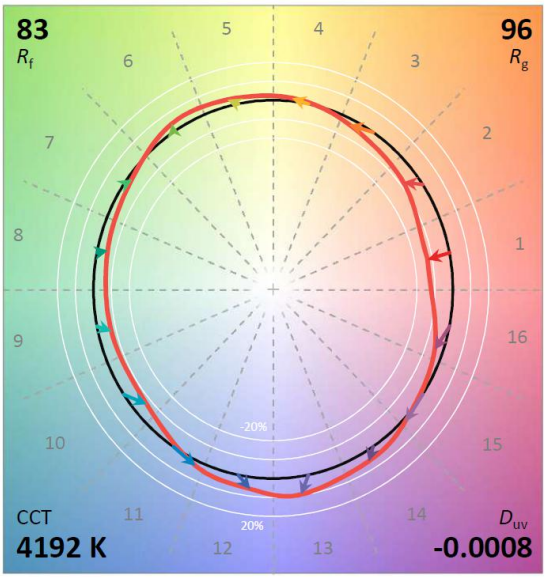


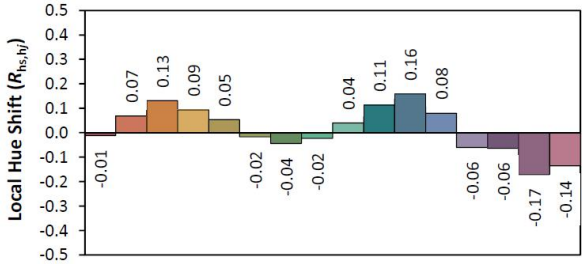
ANSI/IES TM-30-18 Color Rendition Report

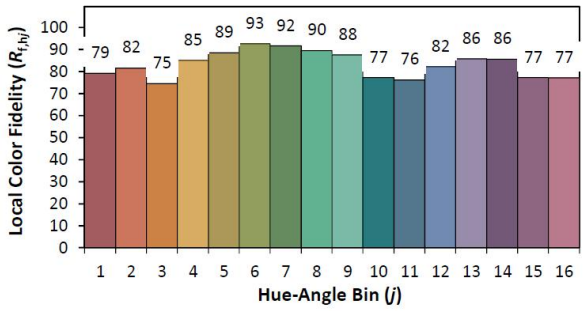
Source:	BL240909002-9	**Manufacturer:**	RZ Lighting Co Ltd
Date:	2024-09-19	**Model:**	WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01, 4000K

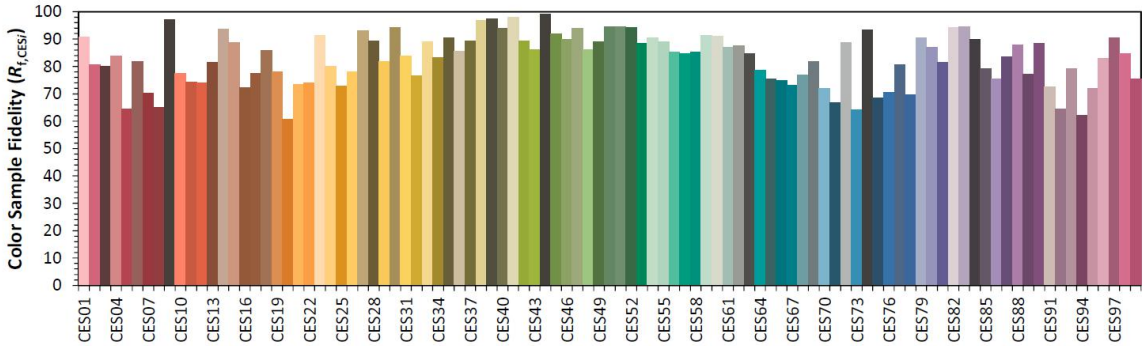












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3719
y	0.3696
u'	0.2223
v'	0.4971

CIE 13.3-1995 (CRI)

R_a	82
R_9	4

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

3.1.2 Model Number: WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01, 5000K

Electrical data

Input Voltage(V)	Frequency (Hz)	Input Current (A)	Power (W)	Power Factor
120.01	60	0.956	114.49	0.998

Photometric data

Luminous Flux (lm)	Efficacy (lm/W)	CCT (K)
18719.18	163.5	4940

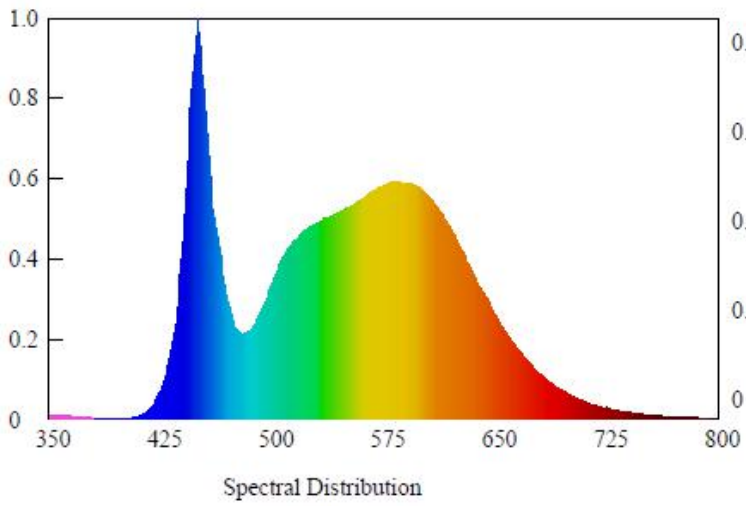
Chromaticity Coordinate

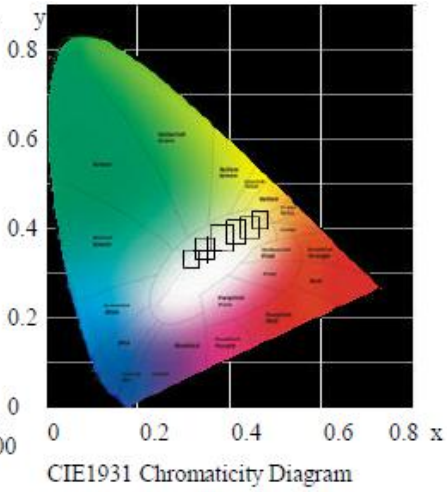
Duv	x	y	u'	v'
-0.00077	0.3467	0.3514	0.2126	0.4848

Color Rendering

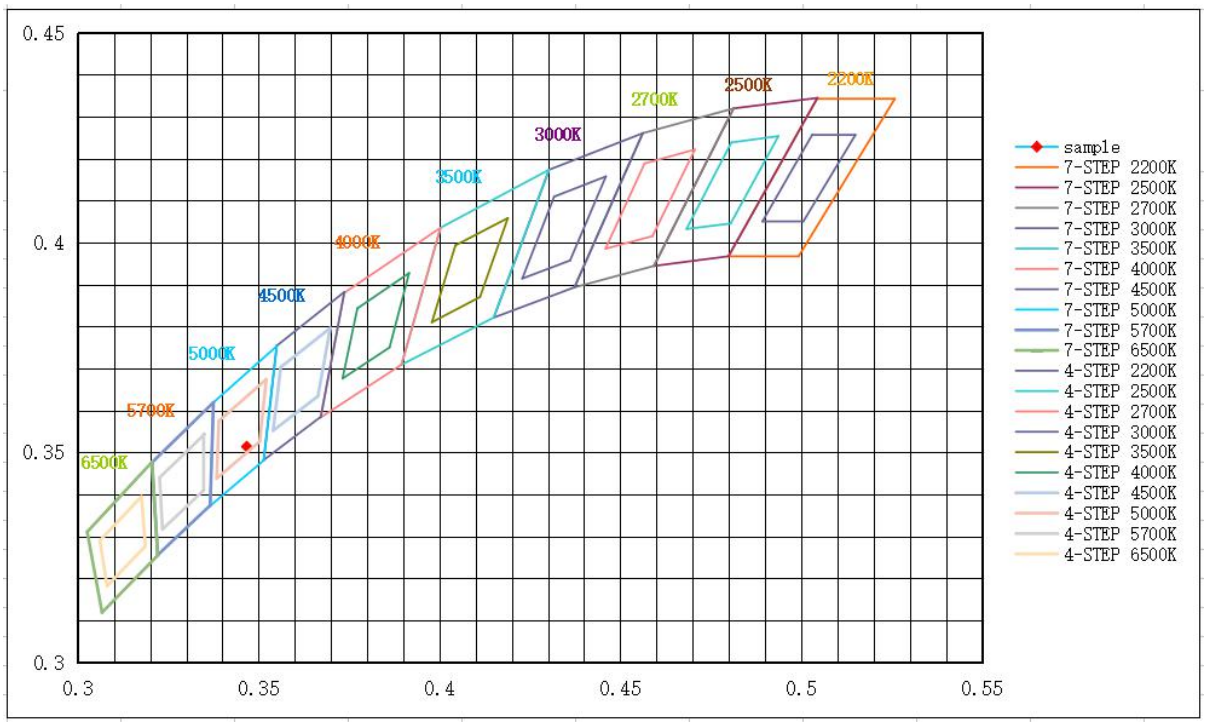
CRI	R9	Rf	Rg	Rcs,h1(%)
83.2	9	83	97	-12

Spectral Distribution



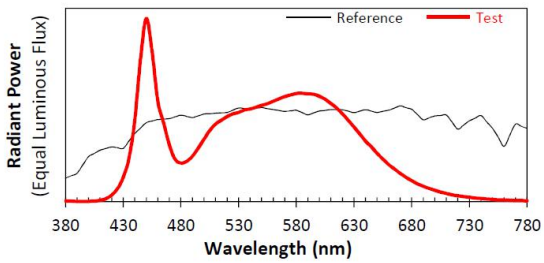


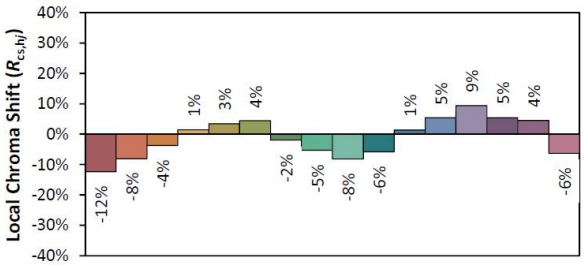
7/4 Step Quadrangle

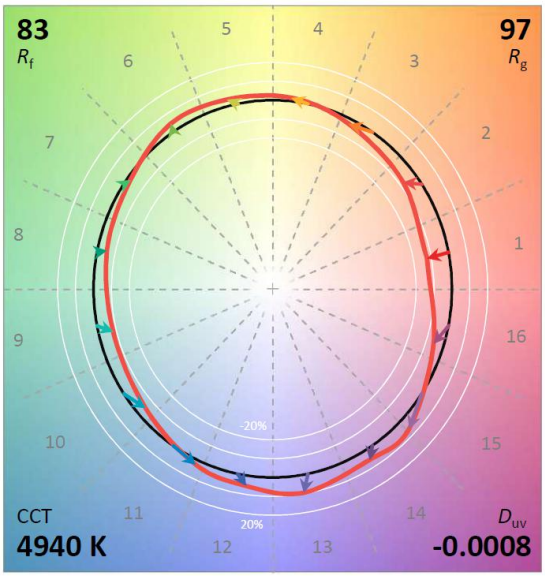


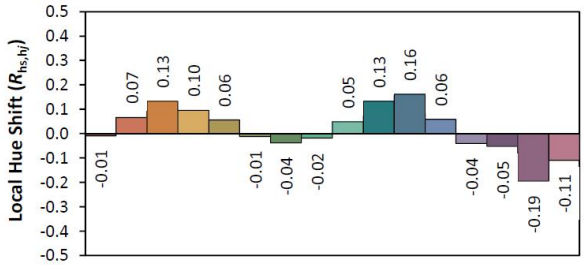
ANSI/IES TM-30-18 Color Rendition Report

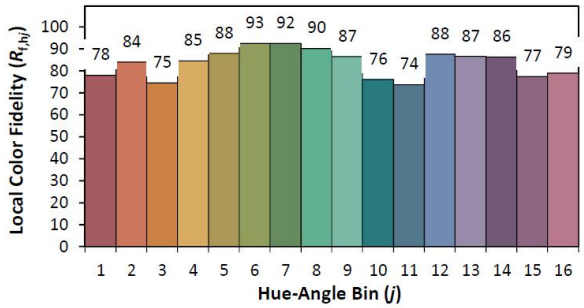
Source:	BL240909002-9	**Manufacturer:**	RZ Lighting Co Ltd
Date:	2024-09-19	**Model:**	WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01, 5000K

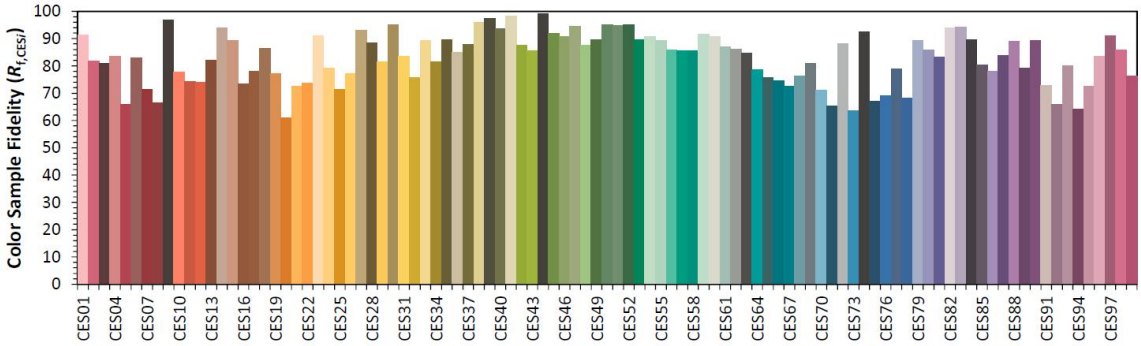












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3467
y	0.3513
u'	0.2126
v'	0.4848

CIE 13.3-1995 (CRI)

R_a	83
R_9	9

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

3.1.3 Model Number: WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01, 5700K

Electrical data

Input Voltage(V)	Frequency (Hz)	Input Current (A)	Power (W)	Power Factor
120.04	60	0.978	117.11	0.998

Photometric data

Luminous Flux (lm)	Efficacy (lm/W)	CCT (K)
18479.54	157.8	5890

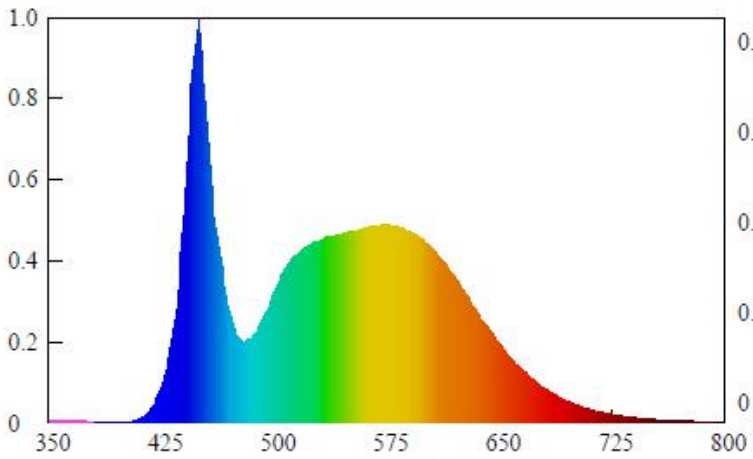
Chromaticity Coordinate

Duv	x	y	u'	v'
+0.0009	0.3241	0.3354	0.2033	0.4734

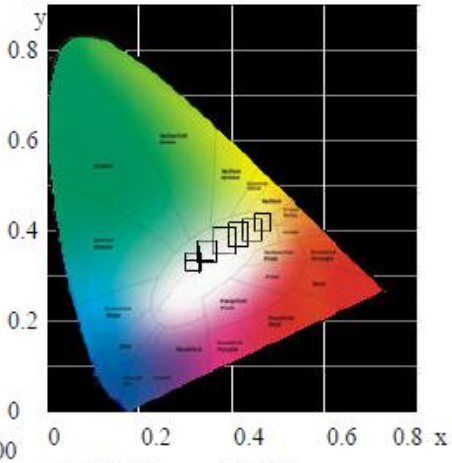
Color Rendering

CRI	R9	Rf	Rg	Rcs,h1(%)
82.8	8	83	97	-13

Spectral Distribution

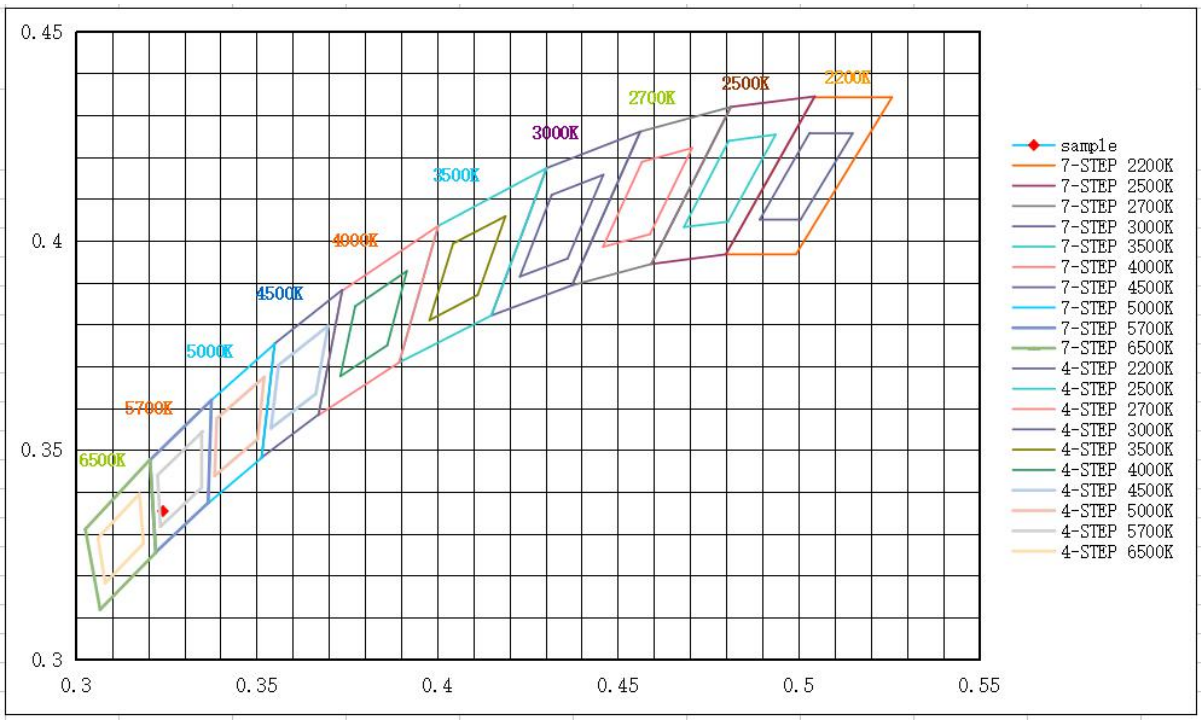


Spectral Distribution



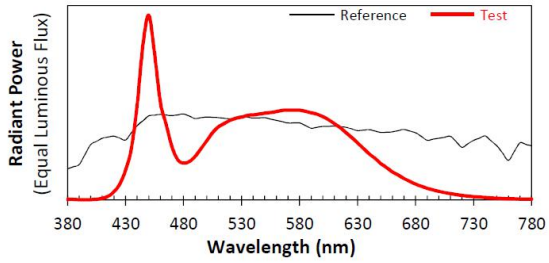
CIE1931 Chromaticity Diagram

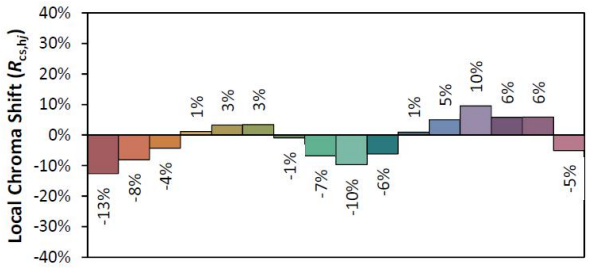
7/4 Step Quadrangle

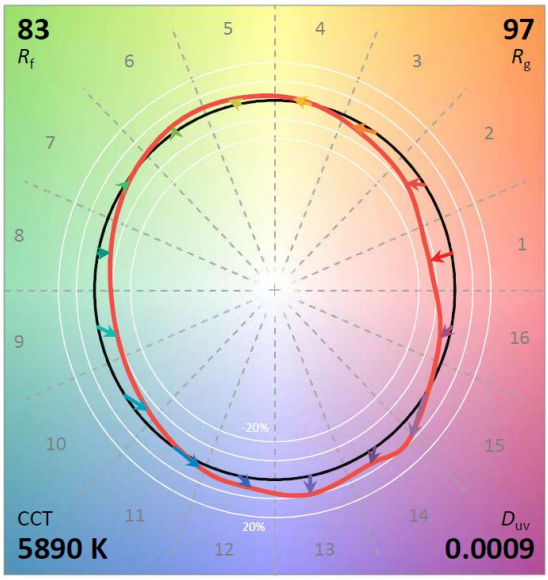


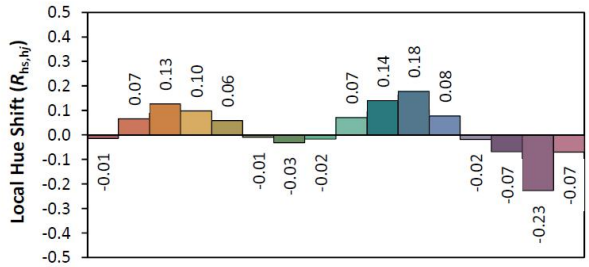
ANSI/IES TM-30-18 Color Rendition Report

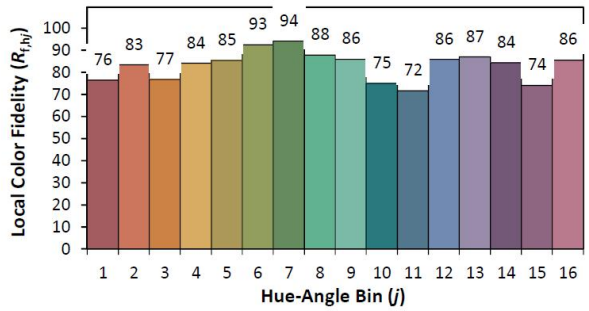
Source:	BL240909002-9	**Manufacturer:**	RZ Lighting Co Ltd
Date:	2024-09-19	**Model:**	WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01, 5700K

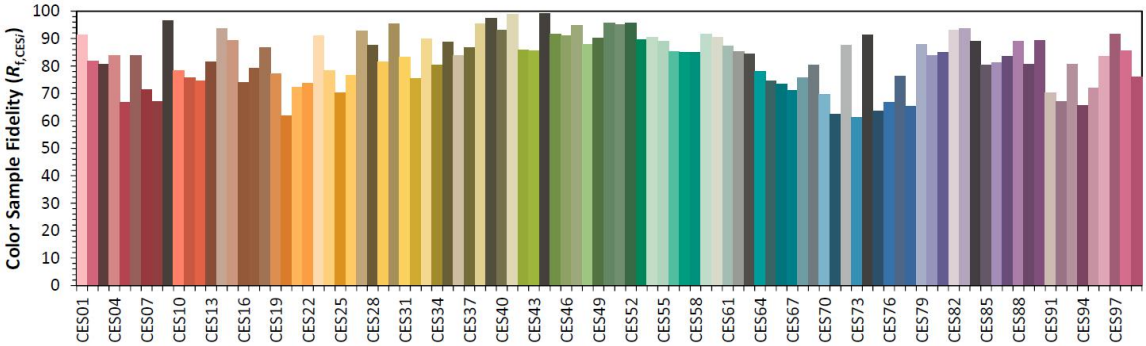












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3241
y	0.3354
u'	0.2033
v'	0.4734

CIE 13.3-1995 (CRI)

R_a	83
R_9	8

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

3.2 Goniophotometer System (Total operating time for luminous intensity distribution: 1.0 hour)

3.2.1 Model Number: WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01

Electrical data

Input Voltage(V)	Frequency (Hz)	Input Current (A)	Power (W)	Power Factor
120.02	60	0.9730	116.63	0.9986

Photometric data

Luminous Flux (lm)	Efficacy (lm/W) (0-90°zone)	Zonal Lumen in 80-90° (%lm)
18030.52	145.86	5.52

IESNA Luminaire Flux Distribution Table:

	Lumens	% Luminaire
FL - Front-Low (0-30)	2565.1	14.2
FM - Front-Medium (30-60)	5695.8	31.6
FH - Front-High (60-80)	3212.0	17.8
FVH - Front-Very High (80-90)	885.5	4.9
BL - Back-Low (0-30)	2007.2	11.1
BM - Back-Medium (30-60)	2125.5	11.8
BH - Back-High (60-80)	467.3	2.6
BVH - Back-Very High (80-90)	52.8	0.3
UL - Uplight-Low (90-100)	475.9	2.6
UH - Uplight-High (100-180)	543.4	3.0
Total	18030.5	100.0

BUG Rating	B3-U4-G5

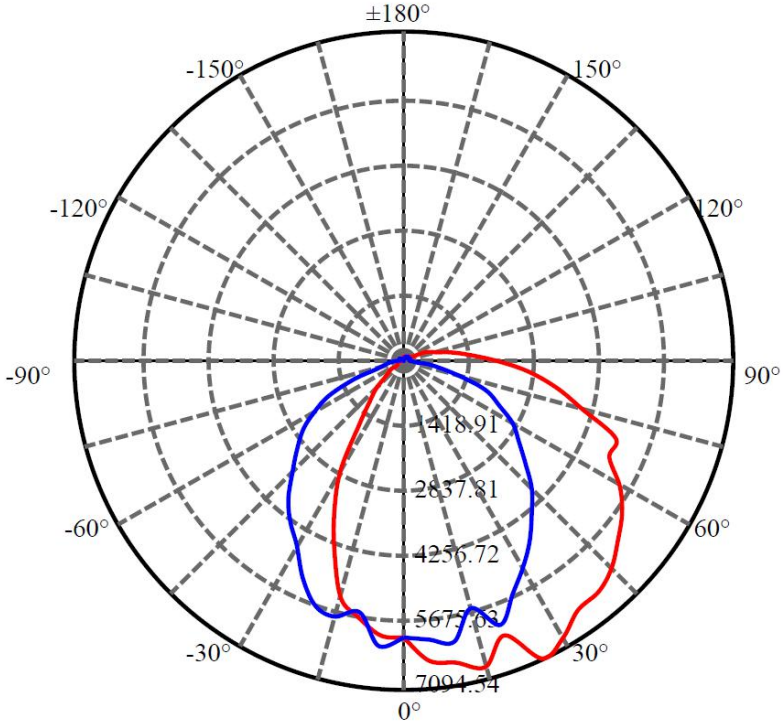
Zonal Flux Diagram

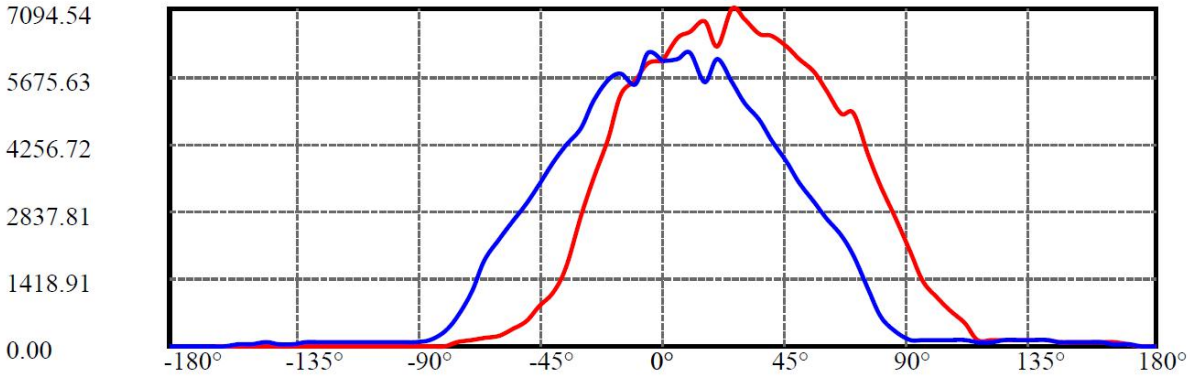
Zonal flux distribution table

γ(°)	Average I(cd)	Zonal F(lm)	Sum F(lm)	Eff Flux(%)	Eff Sum(%)
0.0	5985.830	0.000	0	0.00%	0.00%
5.0	5890.692	141.980	141.98	0.00%	0.79%
10.0	5951.638	423.637	565.618	0.00%	3.14%
15.0	5664.324	689.051	1254.669	0.00%	6.96%
20.0	5470.570	917.672	2172.341	0.00%	12.05%
25.0	5256.501	1125.072	3297.412	0.00%	18.29%
30.0	4817.999	1274.936	4572.349	0.00%	25.36%
35.0	4321.619	1345.873	5918.222	0.00%	32.82%
40.0	3818.357	1358.094	7276.315	0.00%	40.36%
45.0	3473.539	1350.154	8626.469	0.00%	47.84%
50.0	3067.278	1321.668	9948.137	0.00%	55.17%
55.0	2708.654	1255.880	11204.017	0.00%	62.14%
60.0	2438.063	1189.649	12393.666	0.00%	68.74%
65.0	2127.918	1109.999	13503.664	0.00%	74.89%
70.0	1830.806	1002.375	14506.039	0.00%	80.45%
75.0	1482.229	865.974	15372.013	0.00%	85.26%
80.0	1137.163	700.876	16072.89	0.00%	89.14%
85.0	853.456	540.899	16613.789	0.00%	92.14%
90.0	597.928	397.400	17011.189	0.00%	94.35%
95.0	415.133	277.384	17288.573	0.00%	95.89%
100.0	315.451	198.517	17487.09	0.00%	96.99%
105.0	238.419	148.200	17635.29	0.00%	97.81%
110.0	137.189	98.178	17733.468	0.00%	98.35%
115.0	80.141	55.029	17788.498	0.00%	98.66%
120.0	85.448	40.255	17828.753	0.00%	98.88%
125.0	86.733	39.799	17868.552	0.00%	99.10%
130.0	80.031	36.260	17904.812	0.00%	99.30%
135.0	69.983	30.312	17935.124	0.00%	99.47%
140.0	64.152	24.836	17959.961	0.00%	99.61%
145.0	57.795	20.346	17980.306	0.00%	99.72%
150.0	53.110	16.331	17996.638	0.00%	99.81%
155.0	48.604	12.872	18009.51	0.00%	99.88%
160.0	44.168	9.730	18019.24	0.00%	99.94%
165.0	33.914	6.435	18025.675	0.00%	99.97%
170.0	20.592	3.233	18028.908	0.00%	99.99%
175.0	14.704	1.263	18030.171	0.00%	100.00%
180.0	14.268	0.346	18030.517	0.00%	100.00%

Luminous Intensity Distribution Diagram

Light Distribution Curve [Unit:cd]





C0/C180:

C90/C270:

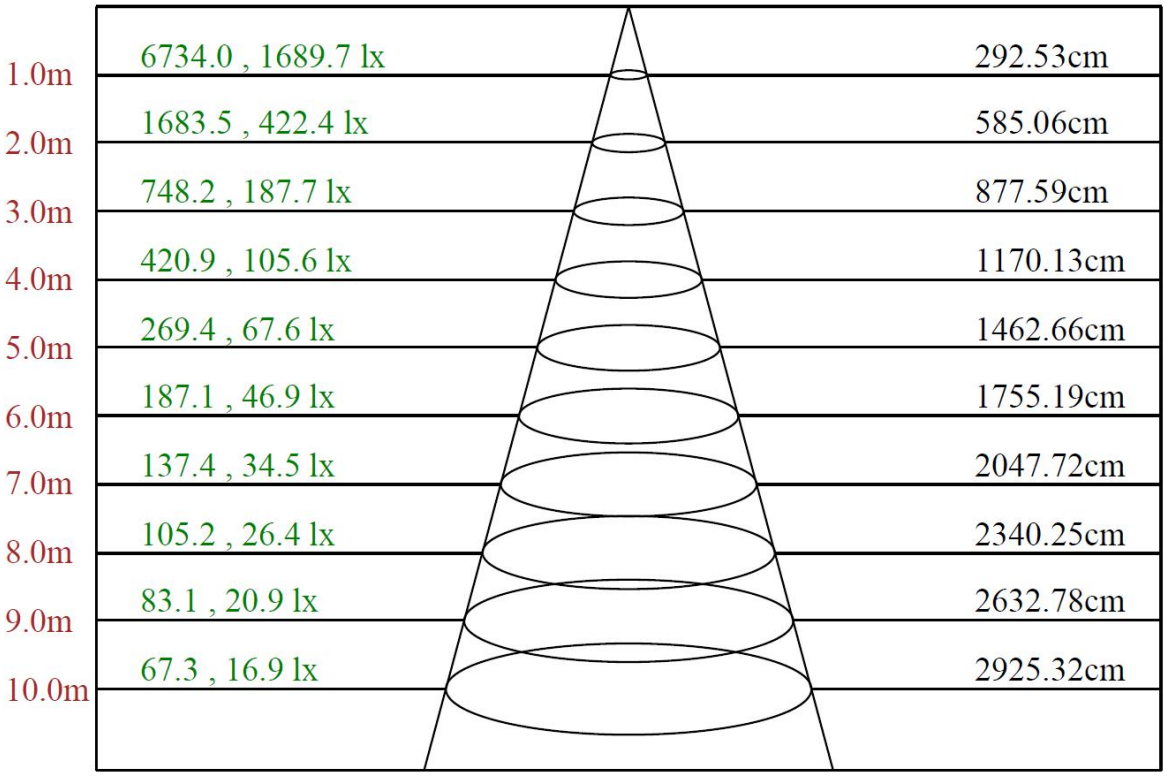
Field angle(10%Imax):C0/180Left:48.9 Right:108.1

:C90/270Left:75.3 Right:80.3

Beam Angle(50%Imax):C0/180Left:28.3 Right:82.6

:C90/270Left:50.0 Right:55.9

Lux distance Curve



Max , Ave Beam angle of C0 plane 111.28

Luminous Intensity Distribution Data

C/γ(°)	0.0	5.0	10.0	15.0	20.0	25.0	30.0	35.0	40.0
0.0	5985.83	6464.35	6627.98	6851.31	6298.52	7094.54	6886.69	6570.49	6504.15
22.5	5985.83	5670.54	6886.69	5975.68	6778.34	6665.57	6212.28	6050.86	5878.39
45.0	5985.83	6342.74	6426.76	5953.57	5889.45	6026.54	6296.30	6072.98	5582.10
67.5	5985.83	5358.77	6327.26	5624.11	5865.13	5756.78	5533.45	5274.74	4706.47
90.0	5985.83	6050.86	6163.63	5542.30	6046.44	5531.24	5111.12	4744.06	4319.52
112.5	5985.83	5971.26	5360.98	5630.74	5378.67	5122.17	4679.94	4235.49	3545.61
135.0	5985.83	5798.79	6260.93	5632.95	4847.99	4514.10	3866.23	3125.49	2283.03
157.5	5985.83	5433.95	5646.22	4777.23	4569.38	4111.67	2891.10	2064.13	1385.30
180.0	5985.83	5935.88	5597.57	5265.90	4381.43	3532.34	2723.06	1668.33	1128.80
202.5	5985.83	5522.40	5210.62	5215.04	4151.47	3881.71	2840.25	1933.67	1279.16
225.0	5985.83	5889.45	5445.00	4746.27	4640.14	4109.46	3481.49	2678.83	1962.41
247.5	5985.83	5051.42	5849.65	4965.18	4854.62	4604.76	4206.75	3627.42	2913.22
270.0	5985.83	6148.16	5513.55	5730.25	5606.42	5164.19	4580.44	4238.15	3880.82
292.5	5985.83	6145.94	5741.30	5938.09	5679.39	5542.30	5416.26	4912.11	4312.89
315.0	5985.83	5655.07	6238.81	6234.39	6077.40	5911.56	6084.03	5827.54	5259.27
337.5	5985.83	6811.51	5929.25	6546.17	6464.35	6535.11	6278.61	6121.62	6152.58
360.0	5985.83	6464.35	6627.98	6851.31	6298.52	7094.54	6886.69	6570.49	6504.15

C/γ(°)	45.0	50.0	55.0	60.0	65.0	70.0	75.0	80.0	85.0
0.0	6316.20	6033.18	5761.20	5387.51	4905.48	4870.10	3989.17	3349.26	2666.67
22.5	6249.87	5913.77	5467.12	5354.35	4974.03	4400.45	4105.26	3319.63	2748.70
45.0	5376.46	5263.69	4912.11	4635.72	4110.78	3587.62	3045.44	2430.08	1774.91
67.5	4375.46	3723.83	3179.22	2849.53	2508.79	2102.60	1655.28	1310.34	901.05
90.0	3897.19	3446.11	3056.94	2705.37	2318.41	1898.29	1197.35	615.37	323.72
112.5	2793.81	2194.59	1688.23	1230.52	903.48	702.71	492.87	312.88	196.13
135.0	1568.82	1137.65	835.60	570.70	391.38	266.89	168.93	102.16	59.48
157.5	1100.06	729.02	447.32	280.38	185.96	120.73	62.80	15.04	7.74
180.0	843.12	532.67	352.46	228.64	171.59	135.99	66.56	11.72	9.95
202.5	953.68	633.28	405.09	250.53	169.38	111.00	55.50	13.49	9.95
225.0	1321.17	948.59	683.47	453.29	306.47	205.20	130.02	72.53	42.01
247.5	2260.92	1728.03	1332.23	1000.55	806.41	602.32	392.93	239.25	144.39
270.0	3409.62	2991.27	2621.12	2241.68	1785.30	1158.21	616.25	316.64	151.47
292.5	3946.05	3315.43	2862.14	2552.13	2185.08	1836.82	1451.41	1052.52	681.93
315.0	5064.68	4856.83	4404.21	4199.01	3653.07	3180.32	2560.53	2008.85	1492.54
337.5	6099.51	5628.53	5330.02	5069.11	4671.09	4113.66	3725.38	3024.88	2444.67
360.0	6316.20	6033.18	5761.20	5387.51	4905.48	4870.10	3989.17	3349.26	2666.67

C/γ(°)	90.0	95.0	100.0	105.0	110.0	115.0	120.0	125.0	130.0
0.0	1971.92	1376.01	993.26	755.11	503.48	122.06	146.60	150.80	145.94
22.5	2015.70	1285.13	931.57	724.60	374.35	112.33	167.83	166.72	149.03
45.0	1220.12	826.76	655.61	471.64	140.41	208.51	199.45	182.86	175.13
67.5	561.64	379.44	285.24	189.72	149.48	136.87	124.27	154.34	161.19
90.0	153.90	113.88	134.00	151.02	137.54	100.39	109.67	127.58	130.68
112.5	137.54	118.08	107.91	100.61	95.52	47.76	61.47	58.82	47.32
135.0	40.24	31.62	27.42	24.77	24.10	22.78	17.03	17.47	19.24
157.5	7.08	7.52	8.40	9.51	10.61	12.83	13.27	13.27	13.71
180.0	8.85	8.85	9.51	10.61	11.28	13.27	15.92	14.82	15.04
202.5	9.29	9.29	9.73	10.84	11.50	13.71	13.27	13.71	13.71
225.0	28.08	20.12	16.58	14.37	12.83	12.16	12.38	12.83	13.27
247.5	99.72	78.05	63.02	49.97	36.04	21.45	19.24	18.80	17.03
270.0	94.64	88.23	91.76	98.40	76.51	82.48	78.28	74.30	67.66
292.5	459.04	328.80	212.05	150.80	130.68	76.95	55.28	74.74	73.19
315.0	1019.79	757.55	599.23	361.31	144.17	179.99	185.30	148.59	107.68
337.5	1739.31	1212.83	901.94	691.43	336.54	118.74	147.93	158.10	130.68
360.0	1971.92	1376.01	993.26	755.11	503.48	122.06	146.60	150.80	145.94

C/γ(°)	135.0	140.0	145.0	150.0	155.0	160.0	165.0	170.0	175.0
0.0	125.15	114.76	95.97	87.34	88.45	90.66	80.93	33.39	13.93
22.5	133.33	116.75	90.44	79.38	76.95	77.83	68.33	25.43	14.82
45.0	141.29	119.62	112.55	115.87	91.76	62.80	34.72	20.12	15.70
67.5	143.95	124.93	101.71	75.62	46.66	38.25	30.51	22.33	17.03
90.0	124.93	123.16	107.02	95.97	93.98	85.35	57.71	36.93	21.01
112.5	42.01	44.22	45.77	45.99	42.68	41.35	39.36	27.20	18.35
135.0	23.44	15.70	13.49	13.93	14.37	16.14	17.91	17.47	15.48
157.5	14.15	14.82	14.37	14.59	14.82	14.59	13.93	14.37	13.49
180.0	15.70	16.14	15.92	15.26	14.82	14.37	13.93	13.49	13.05
202.5	14.59	15.92	15.48	15.26	15.48	14.59	13.71	13.05	13.27
225.0	13.93	14.59	14.82	15.26	15.04	14.59	14.15	13.49	13.05
247.5	15.70	14.82	15.04	18.13	19.90	15.26	14.15	13.71	13.27
270.0	65.01	65.01	70.54	53.07	33.83	16.58	13.05	13.05	12.83
292.5	65.01	61.47	58.15	44.00	42.90	24.10	13.71	13.27	13.05
315.0	68.55	62.80	61.69	69.65	80.27	82.70	40.02	20.56	13.27
337.5	112.99	101.71	91.76	90.44	85.79	97.51	76.51	31.62	13.71
360.0	125.15	114.76	95.97	87.34	88.45	90.66	80.93	33.39	13.93

C/γ(°)	180.0
0.0	14.27
22.5	14.27
45.0	14.27
67.5	14.27
90.0	14.27
112.5	14.27
135.0	14.27
157.5	14.27
180.0	14.27
202.5	14.27
225.0	14.27
247.5	14.27
270.0	14.27
292.5	14.27
315.0	14.27
337.5	14.27
360.0	14.27

4 Additional Test

Electrical data

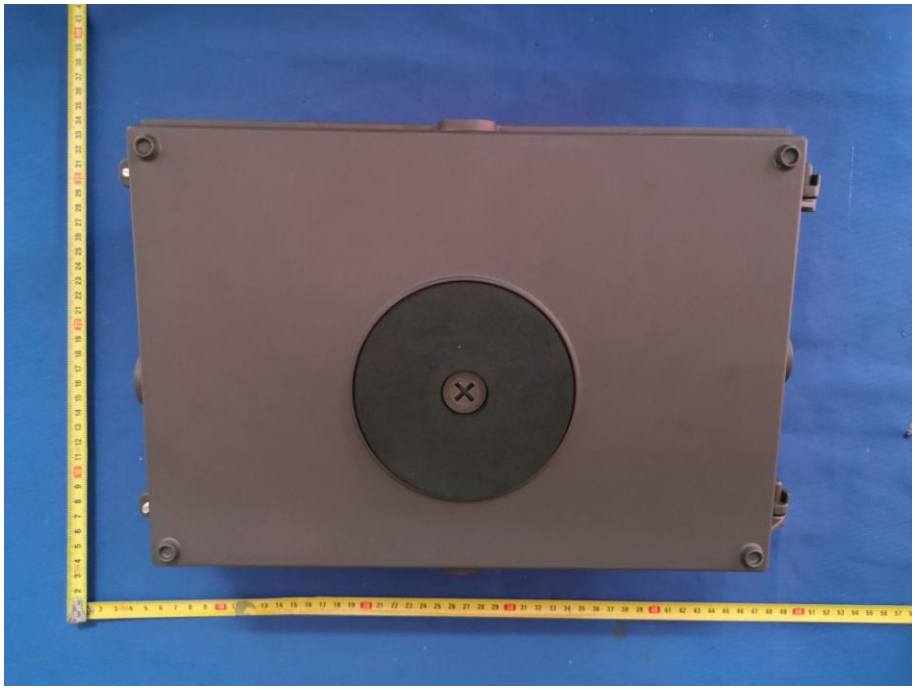
Model Number	CCT(K)	Test Voltage (V)	Frequency (Hz)	Power Factor	THD(%)
WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01	4000	120	60	0.998	3.4
		277	60	0.951	9.6
	5000	277	60	0.952	9.5
	5700	277	60	0.951	9.1

5 Data reporting for white-tunable submissions

(WP09A-14-120L-[Blank;S2;S;S1;E;M;M1;M2][A;D]4C3-01)

ANSI CCT Quadrangle (omit any outside product range) / Worst-Case Value	Actual CCT (K)	Power Consumption (W)	Lumen Output (lm)	Input Control Signal Applied
4000K	4192	116.72	18021.89	0%
5000K	4940	114.49	18719.18	50%
5700K	5890	117.11	18479.54	100%
Lowest Efficacy	4192	116.72	18021.89	0%
Maximum Power	5890	117.11	18479.54	100%

Photo Document



****End of test report****